

MatBYIB: A Matlab-based code for Bayesian inference of extreme mass-ratio inspiral binary with arbitrary eccentricity

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Accurate parameter estimation (PE) of gravitational waves (GW) is essential for GW data analysis. In extreme mass-ratio inspiral binary (EMRI) systems, orbital eccentricity is a critical parameter for PE. However, current software for PE of GW often neglects the direct estimation of orbital eccentricity. To fill this gap, we have developed the **MatBYIB**, a MATLAB-based software package for PE of GW with arbitrary eccentricity. The **MatBYIB** employs the Analytical Kludge (AK) waveform as a computationally efficient signal generator and computes parameter uncertainties via the Fisher Information Matrix (FIM) and the Markov Chain Monte Carlo (MCMC). For Bayesian inference, we implement the Metropolis-Hastings (M-H) algorithm to derive posterior distributions. To guarantee convergence, the Gelman-Rubin convergence criterion (the Potential Scale Reduction Factor $\hat{R}$) is used to determine sampling adequacy, with **MatBYIB** dynamically increasing the sample size until $\hat{R} < 1.05$ for all parameters. Our results demonstrate strong agreement between FIM-based predictions and full MCMC sampling. This program is user-friendly and allows for estimation of gravitational wave parameters with arbitrary eccentricity on standard personal computers. *Code availability:* The implementation is open-source at <https://github.com/GenLiangLi/MatBYIB>.

I. INTRODUCTION

In GW detection and in many other areas, PE is one common part of the statistical analysis, with a goal of inferring the parameters of the system generating the gravitational waves, using data recorded by ground-based detectors LIGO [1–3], Virgo [4], and KAGRA [5], future space-based GW detectors, including LISA [6], Taiji [7], and Tianqin [8], etc. For compact binary coalescence (CBC) detections, in particular with the focus on the study of EMRI, the task of PE is to estimate, as precisely as possible, the masses, spins, sky locations, distance, etc, in order to reveal properties of the astrophysical population, test fundamental physics [9], and to probe possible new physics [10–12].

PE necessitates the construction of precise and rapid GW templates to facilitate swift identification of signals within detector noise. For most LIGO GW sources, the orbital eccentricity is conventionally neglected under the quasi-circular approximation (e.g., Taylor series [13–15]) and these computationally efficient models enable rapid GW parameter estimation. However, orbital eccentricity serves as a critical discriminator for probing the formation environments and mechanisms of compact binary systems [16–24]. Its inclusion proves essential for accurate PE of GW sources [25]. The development of precision templates incorporating eccentricity remains a significant challenge in contemporary GW astronomy.

Current prevalent methodologies for generating eccentric compact binary waveforms include the effective one-body (EOB, SEOBNR) [26], the frequency-domain phenomenological template series (Phenom [27] and IMRPhenomP [28]), eccentric post-circular (EPC) [25]. However, the extreme-mass ratio inspiral binary (EMRI)

with relatively large mass ratios (10^4 to 10^7), requires a longer evolution time and the GW model with higher precision. Currently, the dominant models for EMRI are AK [29], numerical kludge (NK) [30], augmented analytical kludge (AAK) [31] models. Although the AK model exhibits limited accuracy, its computational efficiency and capacity for term extension have enabled widespread application [32].

PE generally requires the use of Bayes' theorem to obtain the posterior probability distribution of parameters. MCMC is the most commonly used method to obtain the posterior probability. There are several prominent, community-developed PE codes, including LALInference [33], PyCBC Inference [34], emcee [35], and Bilby [36]. These packages have been tested by multiple research groups and are widely recognized as standard tools in the field. However, open-access MCMC packages written in MATLAB for GW astronomy observations are not commonly seen.

We have developed a MATLAB-based Bayesian inference toolkit for GW signals from EMRI with arbitrary eccentricities, **MatBYIB**, leveraging the visualization capabilities and concise syntax of MATLAB's graphical user interface. This tool employs the framework of the AK waveform to generate GW. **MatBYIB** derives posterior distributions of GW parameters through MCMC sampling and incorporates the convergence diagnostic method proposed by Gelman and Rubin [37] which defines a Potential Scale Reduction Factor $\hat{R}$, and the MCMC sampling continues until $\hat{R} < 1.05$ [37] for all parameters. The entire program is implemented with multiple independent MCMC chains running in parallel to enhance the computational efficiency and robustness in the sampling process. This design ensures that the toolkit can handle complex and computationally intensive tasks efficiently,

making it suitable for large-scale parameter estimation and predictions. In conclusion, **MatBYIB** provides a simple, efficient, and scalable codes for parameter estimation and predictions for future GW detectors such as LISA.

II. THEROY

A. Waveform generation

A compact stellar-mass object m_2 (typically a stellar black hole (BH) or a neutron star (NS)) orbits around a central supermassive BH of mass m_1 , and this system emits GW in the millihertz frequency band, which is

called EMRI. The AK waveform is currently one of the most widely used GW templates for EMRI. Despite its many limitations, such as low accuracy, it is convenient to use for simple parameter error estimation because of its lower computational complexity, and it can well simulate the motion of particles far from the BH using post-Newtonian approximation.

The AK waveform is mainly composed of two parts. The first part describes the orbital dynamics of the small object (Effective Single-body Approximation assumed that large central BH is stationary [38]) and the second part is waveform production by using the Peters-Mathews waveform equation under the quadrupole approximation [39].

The evolution of GW orbital parameters with time can be expressed as [40]:

$$\frac{d\Phi}{dt} = 2\pi f_{orb}, \quad (1)$$

$$\begin{aligned} \frac{df_{orb}}{dt} = & \left(\frac{96}{10\pi}\right) \left(\frac{c^6 \mu}{G^2 M^3}\right) \left(\frac{2\pi G M f_{orb}}{c^3}\right)^{11/3} (1-e^2)^{-9/2} \left[(1-e^2) \left(1 + \frac{73}{24}e^2 + \frac{37}{96}e^4\right) \right. \\ & + \left(\frac{2\pi G M f_{orb}}{c^3}\right)^{2/3} \left(\frac{1273}{336} - \frac{2561}{224}e^2 - \frac{3885}{128}e^4 - \frac{13147}{5376}e^6\right) \\ & \left. - \left(\frac{2\pi G M f_{orb}}{c^3}\right) \left(\frac{c}{G} S/M^2\right) \cos \lambda (1-e^2)^{-1/2} \left(\frac{73}{12} + \frac{1211}{24}e^2 + \frac{3143}{96}e^4 + \frac{65}{64}e^6\right)\right], \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{de}{dt} = & -\frac{e}{15} \left(\frac{c^3 \mu}{G M^2}\right) \left(\frac{2\pi G M f_{orb}}{c^3}\right)^{8/3} (1-e^2)^{-7/2} \left[(1-e^2)(304 + 121e^2)(1 + 12\left(\frac{2\pi G M f_{orb}}{c^3}\right)^{2/3}) \right. \\ & \left. - \frac{1}{56} \left(\frac{2\pi G M f_{orb}}{c^3}\right)^{2/3} (8 \cdot 16705 + 12 \cdot 9082e^2 - 25211e^4)\right] + e \frac{c^3 \mu}{G M^2} (S/M^2) \cos \lambda \\ & \left(\frac{2\pi G M f_{orb}}{c^3}\right)^{11/3} (1-e^2)^{-4} \left(\frac{1364}{5} + \frac{5032}{15}e^2 + \frac{263}{10}e^4\right), \end{aligned} \quad (3)$$

$$\begin{aligned} \frac{d\tilde{\gamma}}{dt} = & 6\pi f_{orb} \left(\frac{2\pi G M f_{orb}}{c^3}\right)^{2/3} (1-e^2)^{-1} \left(1 + \frac{1}{4} \left(\frac{2\pi G M f_{orb}}{c^3}\right)^{2/3} (1-e^2)^{-1} (26 - 15e^2)\right) \\ & - 12\pi f_{orb} \left(\frac{c}{G} S/M^2\right) \cos \lambda \left(\frac{2\pi G M f_{orb}}{c^3}\right) (1-e^2)^{-3/2}, \end{aligned} \quad (4)$$

$$\frac{d\alpha}{dt} = 4\pi f_{orb} \left(\frac{2\pi G M f_{orb}}{c^3}\right) \left(\frac{c}{G} S/M^2\right) (1-e^2)^{-3/2}. \quad (5)$$

Where, M , Φ , f_{orb} , e are the total mass, orbital phase, frequency and eccentricity, respectively. $\tilde{\gamma}$, α are two precession angles. We can also include higher-order PN terms if necessary.

We establish a Cartesian coordinate system in the ecliptic plane, with the axes denoted as x , y , and z . The spin angular momentum of the BH is represented by the vector $\hat{S}$, where S signifies the magnitude of the angular momentum. Meanwhile, the orbital angular momentum is represented by the vector $\hat{L}(t)$, whose orientation is determined by the inclination angle λ —the angle between the vectors $\hat{L}$ and $\hat{S}$ —and the azimuthal angle $\alpha(t)$. The projection of the orbital angular momentum in the direction of wave propagation can be represented as

$$\hat{L} \cdot \hat{n} = \hat{S} \cdot \hat{n} \cos \lambda + \frac{\cos \theta_S - \hat{S} \cdot \hat{n} \cos \theta_K}{\sin \theta_K} \sin \lambda \cos \alpha + \frac{(\hat{S} \times \hat{z}) \cdot \hat{n}}{\sin \theta_K} \sin \lambda \sin \alpha, \quad (6)$$

where θ_K and ϕ_K are the sky orientation angles of the spin angular momentum vector. If the center BH is the

Schwarzschild BH, the angle between $\tilde{L}$ and $\tilde{S}$ is $\lambda = 0$. For the n -th harmonic wave h_n^+ can be written as

$$\begin{aligned} h_n^+ &= -\frac{1}{D} \{ [1 + (\hat{L} \cdot \hat{n})^2] [a_n \cos(2\xi) - b_n \sin(2\xi)] + [1 - (\hat{L} \cdot \hat{n})^2] c_n \}, \\ h_n^\times &= \frac{2}{D} (\hat{L} \cdot \hat{n}) [b_n \cos(2\xi) + a_n \sin(2\xi)]. \end{aligned} \quad (7)$$

In which, ξ is an azimuthal angle used to measure the direction of the pericenter relative to the x axis [40], and $\hat{x}$ is defined as $\hat{x} \equiv \frac{-\hat{n} + \hat{L}(\hat{L} \cdot \hat{n})}{\sqrt{1 - (\hat{L} \cdot \hat{n})^2}}$. D is the luminosity distance, a_n , b_n , and c_n are the superpositions of Bessel functions related to the eccentricity [40]. As for the circular orbital case, a_n, b_n is zero. Therefore, we have obtained the GW source signal by superposition of harmonic waves.

B. Response functions

To analyze the GW source, it is necessary to transform the GW source into the detector's reference frame, and this transformation necessitates consideration of the detector's response functions. The representation of GWs in the detector frame has been formulated as [41].

$$h(t) = \frac{1}{2} [F^+(t)h^+(t) + F^\times(t)h^\times(t)], \quad (8)$$

where F^+ and $F^\times$ are the response functions.

$$F^+(\theta, \phi, \psi) = \left[\frac{1}{2} (1 + \cos^2 \theta) \cos 2\phi \sin 2\psi - \cos \theta \sin 2\phi \cos 2\psi \right], \quad (9)$$

$$F^\times(\theta, \phi, \psi) = \left[\frac{1}{2} (1 + \cos^2 \theta) \cos 2\phi \sin 2\psi + \cos \theta \sin 2\phi \cos 2\psi \right], \quad (10)$$

where (θ, ϕ) are the sky location and ψ is the polarization angle of the GW source [40]. For space-based detectors, such as LISA and Taiji, the sky angles θ, ϕ, ψ in the detector frame change over time due to the continuous rotation of the detectors [41]. By performing the Fourier transform on Eq. (8), we can obtain the GW in the frequency domain $\tilde{h}(f)$.

C. Fisher Information Matrix

FIM provides a computationally efficient framework for rapid precision estimation of GW parameters by quantifying the local curvature of the likelihood surface in the parameter space. As shown in Table I, the GW from two inspiraling bodies can be described by a set of parameters $= (\theta^1, \dots, \theta^k)$, we consider this set of waveforms as

a multidimensional surface embedded in the vector space of all possible measured signals [42]. The maximum likelihood estimator is indeed the value of θ that provides the highest signal-to-noise ratio (SNR) in the matched filtering [43]. To compute the likelihood function, and hence the posterior probability, we assume for simplicity that the noise $n(t)$ is stationary and Gaussian. Given the detected signal $s(t) = h(t; \theta) + n(t)$, where $h(t; \theta)$ is the GW template depending on parameters and $n(t)$ is the detector noise, the likelihood $\Lambda(s | \theta)$ is:

$$\Lambda(s | \theta) \propto \exp \left[-\frac{1}{2} (s - h(\theta) | s - h(\theta)) \right], \quad (11)$$

The inner product above is defined as $(n | n) = \text{Re} \int_{-\infty}^{\infty} df \frac{\tilde{n}^*(f) \tilde{n}(f)}{(1/2) S_n(f)}$ and S_n is the sensitive curve of the detector.

In the Bayesian approach

$$P(\theta | s) = \frac{P^{(0)}(\theta) \Lambda(s | \theta)}{P(s)}, \quad (12)$$

where $P(\theta | s)$ is posterior distribution and $P^{(0)}(\theta)$ is prior distribution and $P(s)$ is the evidence. We can expand the $(s - h | s - h)$ around the maximum likelihood estimator θ_{ML} and the linear term of the expansion vanishes. We can get [43]

$$P((\theta | s) \propto \exp \left\{ -\frac{1}{2} \Gamma_{ij} \Delta \theta^i \Delta \theta^j \right\}, \quad (13)$$

where $\Gamma_{ij} = (\partial_i \partial_j h | h - s) + (\partial_i h | \partial_j h)$. In the high SNR (e.g., $\text{SNR} > 10$ [42]) regime, $(h - s) \ll h$ and

$$\Gamma_{ij} = (\partial_i h | \partial_j h), \quad (14)$$

the Γ is FIM and the covariance matrix is given by $\langle \Delta \theta^i \Delta \theta^j \rangle = (\Gamma^{-1})^{ij}$ and the expected value of parameter error $\Delta \theta^i$ is given by $\Delta \theta^i = \sqrt{\langle \Delta \theta^i \Delta \theta^i \rangle} = \sqrt{(\Gamma^{-1})^{ii}}$.

D. Markov Chain Monte Carlo

The MCMC method is a technique for sampling from complex probability distributions by constructing a Markov chain. The core thought is to design a Markov chain whose samples converge to the target distribution. Various sampling methods have been developed within

the MCMC framework, including M-H [44, 45], No-U-Turn sampling [46], Gibbs sampling [47–49], Blocked M-H [50], Parallel Tempering [51–53], and Reversible Jump, among others. In this study, we employ the M-H sampling method. The process begins by selecting a random position in the parameter space. For the $(i+1)$ -th step, the position of θ_{i+1} is determined only based on the previous point, θ_i . Typically, this is achieved by constructing a transition matrix $q(\theta_{i+1} | \theta_i)$ and we set the transition matrix satisfy the Normal distribution [54]

$$q(\theta_{i+1} | \theta_i) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(\theta_{i+1}-\theta_i)^2}{2\sigma^2}}, \quad (15)$$

where σ is a covariance matrix, which we wish to be as small as possible for efficiency purposes. A candidate point is randomly drawn by the transition matrix $q(\theta_{i+1} | \theta_i)$. then, we need to calculate the acceptance probability $c(i+1, i)$:

$$c(i+1, i) = \min\{P(\theta_{i+1})q(\theta_{i+1}|\theta_i) - P(\theta_i)q(\theta_i|\theta_{i+1}), 0\}. \quad (16)$$

Where the $P(\theta_{i+1})$ is the poster distribution obtained from Eq. (12). To make the Markov chain converge more quickly, a random value $u \sim \text{uniform}(0, 1)$ is typically generated [45]. If $\log(u) < c(i+1, i)$, then retain $P(\theta_{i+1})$, set $P(\theta_i) = P(\theta_{i+1})$, and update. If $\log(u) > c(i+1, i)$, then retain $P(\theta_i)$ and repeat the process. The pseudocode of the M-H algorithm is given in Algorithm 1.

Algorithm 1: Pseudo-code of Metropolis-Hastings

Input: Initialize the Markov chain state θ , the number of iterations N

Output: The list of samples

while $i \leq N$ **do**

 Calculate the target distribution $P(\theta_i)$ at the current state using Eq. (12);

 Generate a proposal θ_{i+1} by the transition kernel $q(\theta_{i+1} | \theta_i)$;

 Compute the acceptance probability $c(i+1, i)$ using Eq. (16);

 Generate a uniform random number $u \sim \text{uniform}(0, 1)$;

if $\log(u) < c(i+1, i)$ **then**

 Accept the proposal: $\theta_i = \theta_{i+1}$;

end

$i \leftarrow i + 1$;

end

E. Convergence Diagnostics

We use the Gelman-Rubin method [37] to optimize the convergence criteria of MCMC. After discarding the burn-in period of the Markov chains, each chain is split

into two parts, we assume that there are $2k$ Markov chains, each containing n iterations, and the value of the i -th iteration of the j -th chain for the target parameter θ is denoted as $\theta_{i,j}$. The variance between chains B reflects the differences between the means of different chains.

$$B = \frac{n}{2k-1} \sum_{j=1}^{2k} (\bar{\theta}_j - \bar{\theta})^2, \quad (17)$$

where $\bar{\theta}_j$ is the sample mean of the j -th chain, and $\bar{\theta}$ is the overall mean. The within-chain variance (W) characterizes the variability within a single chain:

$$W = \frac{1}{2k} \sum_{j=1}^{2k} \left(\frac{1}{n-1} \sum_{i=1}^n (\theta_{i,j} - \bar{\theta}_j)^2 \right). \quad (18)$$

By combining B and W through weighted integration, we obtain the variance estimate for the target parameter θ :

$$\text{var}(\theta) = \left(\frac{n-1}{n} \right) W + \left(\frac{1}{n} \right) B. \quad (19)$$

Under stationarity or as $n \rightarrow \infty$, $\text{var}(\theta)$ is an unbiased estimate and approximately equal to W . The Potential Scale Reduction Factors is defined as:

$$\hat{R} = \sqrt{\frac{\text{var}(\theta)}{W}}, \quad (20)$$

when $\hat{R} \approx 1$, the chain has reached stationarity; when $\hat{R} > 1.05$ [37], the chain has not converged, and the number of iterations n needs to be increased. We set the program to automatically add the value of n and continue sampling.

III. SOFTWARE ARCHITECTURE

As shown in Fig. 1, the software is divided into five major modules, namely the common constants module (**Common_CF.m**), the waveform generation module (**Waveform.m**), the detector module (**Detector.m**), the FIM module (**Fisher_Matrix.m**), and the MCMC module (**MCMC.m**). The common constants module (**Common_CF.m**) primarily encompasses common constants (**Common_constants.m**) and the input parameter function (**Readinput()**). The waveform generation module (**Waveform.m**) provides GW waveform templates, and the detector module offers functions related to the detectors, such as the detector response functions, noise function and sensitivity curves. Using the FIM module, we can initially calculate the SNR of GW and the covariance of parameters, ultimately yielding parameter errors. Furthermore, the posterior distribution of the parameters can be obtained through the **MCMC_run()** function within the MCMC module (**MCMC.m**).

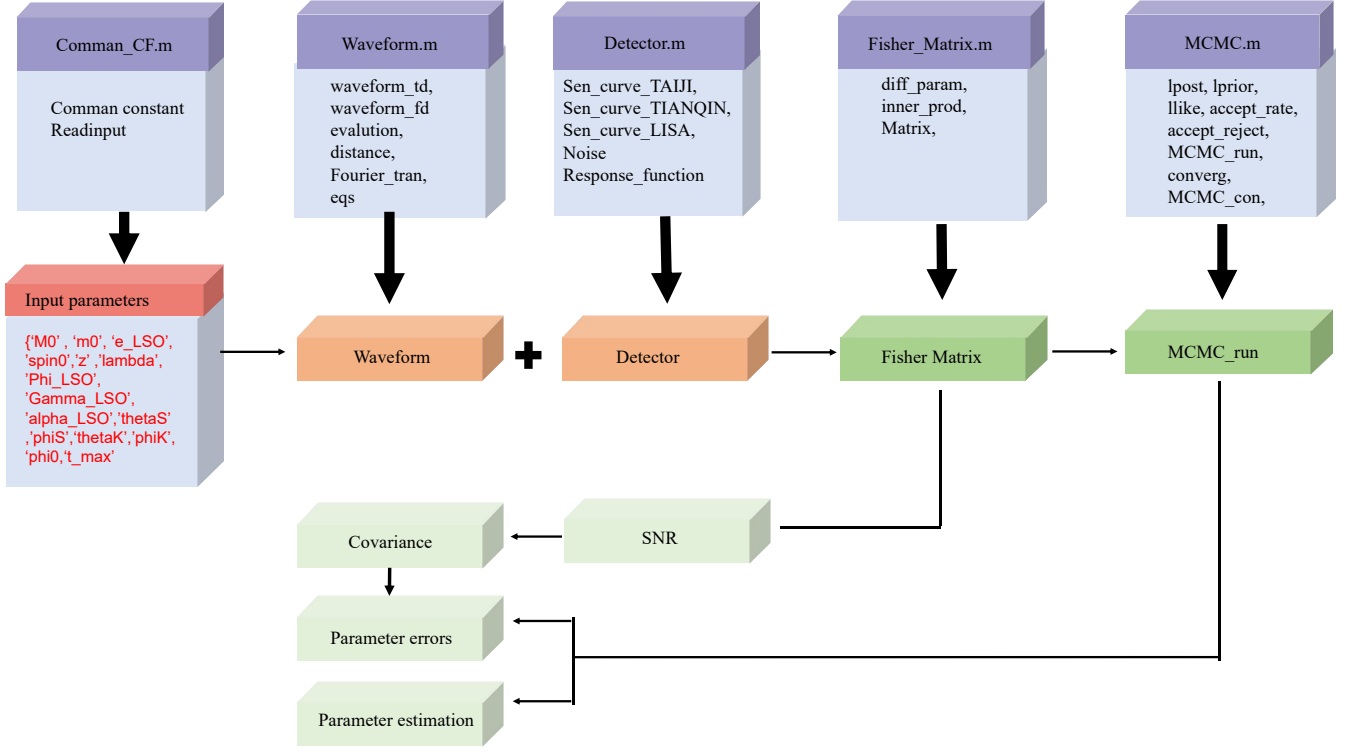


FIG. 1. Schematic Diagram of the Software Architecture

Common_CF.m: This module defines the values of physical constants and the function to read input files (`Readinput()`). Users can fill in the corresponding parameters as needed for their models.

Waveform.m: This module defines a set of orbital evolution functions (see Eq. (5)) and generates waveforms according to Eq. (7). The main functions are as follows:
eqs(): Defines a set of ordinary differential equations (ODEs) Eq. (5) and returns the values of the orbital parameters as they evolve. Note that there may be slight differences in the ODE solvers due to the different versions of MATLAB.
evolution(): Calculate the solutions of the kludge ODE equations defined in `eqs()`. To solve the ODE system, we utilize the `ODE45` function in MATLAB. This choice facilitates the switch between different native solvers available in the library.

get_Aplus_A_cross(): It calls `evolution()` and calculates the intensity of different-order harmonic GW in the source coordinate system according to the Eq. (7).
waveform_td(): It calls `eqs()`, `evolution()` and `get_Aplus_A_cross()`, to compute the time-domain GW including the detector response function from the detector module, and then performs its Fourier transform via the function. `Fourier_tran()` in `Common_CF.m`. This process is integrated into the functions `Fisher_Matrix()` and `MCMC_run()` detailed below.

Detector.m: This module defines the sensitivity curves of various GW detectors, including LISA [55], Taiji [7] and Tianqin [8], as well as their response functions. Users can modify the detector sensitivity curve directly in the main.m file. For example, they can call the `Sen_curve_LISA()` function from the `Detector.m` module to obtain the LISA sensitivity curve.

get_noise(): This defines the detector noise function, and the noise satisfies the Gaussian distribution.

Fisher_Matrix.m: This module defines functions related to the calculation of the FIM, including:

Diff_param(): This function defines the partial derivatives of the GW signal with respect to different parameters. It generates the frequency domain function by calling the `waveform_fd()` and uses the central difference to obtain the partial derivatives of the GW signal with respect to different parameters.

Matrix(): It calls `diff_param()` and `inv()` functions, the latter being MATLAB's built-in function for calculating the inverse of a matrix.

MCMC.m: This module consists of functions related to MCMC. The functions mainly include:

lprior(), **lpost()**: These functions define the prior distribution and the posterior probability, respectively. They can be manually set to specific ranges.

llike(): This function computes the likelihood according

to Eq. (11). It is used in `MCMC_run()` to compute the likelihood at each step.

accept_reject(): This function corresponds to Eq. (16).

MCMC_run(): It calls `waveform_fd()` to generate the GW signal and calculates the posterior distribution at each step according to Eq. (11), and calls `accept_reject()` to determine whether to retain the current particle's posterior distribution value, Eq. (16), then continues to iterate.

converg(): The function uses Eq. (17), Eq. (18) and Eq. (19) to assess convergence of sampling. If $\hat{R} > 1.05$, the chain is considered non-convergent, and we will increase the sampling points. Typically, in our program, the number of points is increased to 1.25 times the original total number, after which `MCMC_contin()` is invoked to resume sampling.

MCMC_contin(): It's usage is identical to that of `MCMC_run()`.

IV. TEST AND NUMERICAL EXAMPLES

The implementation details, input parameter specifications, and considerations concerning numerical precision are addressed in the accompanying source code and program documentation. Therefore, this section presents only selected numerical examples for demonstration purposes.

A. GW waveform

Upon the execution of the `main.m` file, the **Common_CF** module is activated, which facilitates the automatic retrieval of binary system parameters from the 'input.txt' file (see Table I).

Following the loading of the waveform module, the `evaluation()` function is invoked, enabling a comprehensive analysis of the variations in the evolutionary orbital parameters of the binary system.

Fig. 2 shows the variation of the orbital parameters of the system: frequency, eccentricity over time. one can observe that with the accumulation of duration time, the orbital frequency progressively increases. Conversely, due to the influence of GW radiation emitted by the system, the orbital eccentricity diminishes over time. We can also demonstrate the variation of the eccentricity with frequency.

We executed the `waveform()` function, which automatically selects the response function in **Detector.m** model based on provided detector selected in 'input.txt', and get the GW in time-domain (As shown in Fig. 3). Subsequently, applying a Fourier transform (`Fourier_tran()`) allows us to derive the frequency-domain waveform and the characteristic strain of the GW (As shown in Fig. 4).

B. Fisher Information Matrix

Before the execution of the **Fisher_matrix.m**, it is imperative to first calculate the SNR of the GW signal. The FIM can be used with greater accuracy only when the SNR is sufficiently high. By integrating the characteristic frequency spectrum of the GW, we can obtain their SNR.

By executing the `diff_param()` function, we can obtain the partial derivatives of the GW signal for various parameters. Utilizing the `inner_prod()` function for the convolution between signals, and applying Eq. (14) to derive the Γ , we can then invoke the built-in MATLAB function `inv()` to acquire the covariance matrix for the different parameters.

In Fig. 5, we plot the results of the posterior distributions for the central BH, the small body mass, the eccentricity corresponding to the last stable orbit, and the spin of BH. These results were obtained using 5000 samples, with the covariance matrix derived from the FIM. The red dashed line represents the true value, while the thin blue dashed lines indicate the 1σ confidence interval.

C. MCMC

For parameters requiring Bayesian posterior estimation, we first define their prior probability distributions. The posterior probabilities are then computed using the likelihood function `lpost()`. We initialize multiple parallel Markov chains (typically set to $2n + 1$ chains, where n represents the dimensionality of parameters needing estimation). The sampling process is implemented through the `MCMC_run()` function following the M-H algorithm (Eq. (16)). Convergence is monitored by calculating the Potential Scale Reduction Factors $\hat{R}$ via the `converge()` function. If $\hat{R} > 1.05$ (indicating non-convergence), we increase the sampling iterations and resume the process using `MCMC_contin()` until all $\hat{R} < 1.05$ criteria are satisfied.

In Fig. 6, we present the posterior distributions for various parameters ($m_1, m_2, e_{\text{LSO}}, S$). The blue dashed line indicates the median value, while the thin blue lines denote the credible intervals 68% derived from the posterior distribution and the red dashed line represents the true parameter values. Our analysis reveals that the results estimated via MCMC are in good agreement with those obtained from the FIM in Fig. 5. Although the MCMC-derived posteriors exhibit marginally broader confidence intervals, they reflect a more complete exploration of the parameter space.

As shown in Table II, the diagnostic Potential Scale Reduction Factors ($\hat{R}$) confirm the convergence of our MCMC chains after 10,000 iterations, with all parameters achieving $\hat{R} < 1.05$. This convergence behavior indicates that our posterior distributions have properly sampled the target parameter space.

TABLE I. Summary of physical parameters and their meaning. The angles (θ_S, ϕ_S) and (θ_K, ϕ_K) are associated with a spherical coordinate system attached to the ecliptic. $\hat{L}$ and $\hat{S}$ are unit vectors in the directions of the orbital angular momentum and the MBH's spin, respectively.

line number	Values	Units	Parameters	physical quantity
1	$1.e+6$	$M_\odot$	m_1	the mass of central BH
2	10	$M_\odot$	m_2	the mass of rotating object
3	0.3	--	e_{LSO}	where e_{LSO} is the last stable circular orbital eccentricity
4	0.01	--	S/M^2	magnitude of spin angular momentum of MBH
5	0.01	--	z	red shift
6	60	$^\circ$	λ	$\cos\lambda = \hat{L} \cdot \hat{S}$
7	0	--	ϕ_{LSO}	where ϕ_{LSO} is the last stable circular orbital mean anomaly
8	60	$^\circ$	γ_{LSO}	where γ_{LSO} is the angle (in orbital plane) between $\hat{L} \times \hat{S}$ and pericenter
9	60	$^\circ$	α_{LSO}	where α_{LSO} is the azimuthal direction of $\hat{L}$ in the orbital plane
10	60	$^\circ$	θ_S	the source direction's polar angle
11	60	$^\circ$	ϕ_S	azimuthal direction to source
12	60	$^\circ$	θ_k	the polar angle of MBH's spin
13	60	$^\circ$	ϕ_k	azimuthal direction of MBH's spin
14	0	$^\circ$	ϕ_0	The mass of rotating object
15	$3.14e+6$	s	t_c	t_c is time where orbit is last stable circular orbit

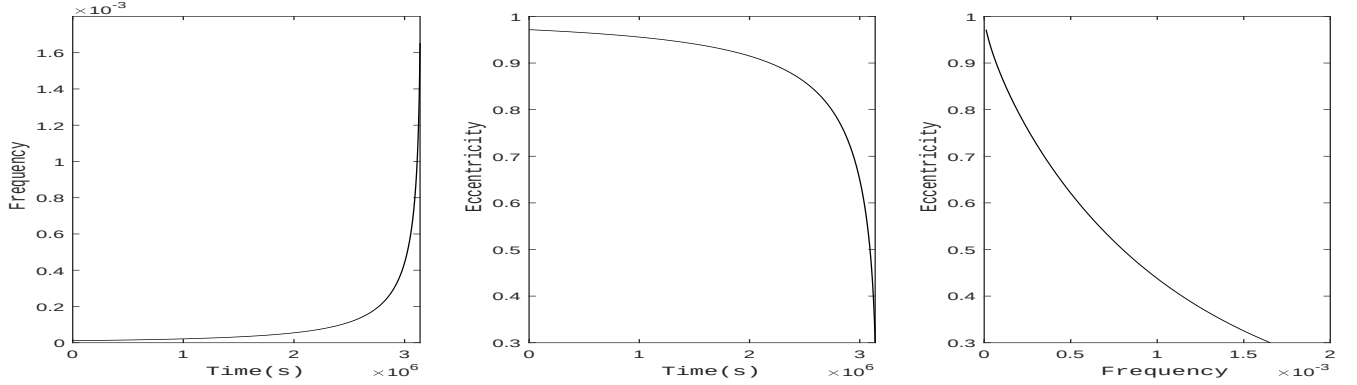


FIG. 2. The evolution of orbital frequency (Left) and eccentricity (Middle) with time, and the evolution of orbital eccentricity as a function of frequency (right). The system parameters are as follows: the mass of the central BH is $M = 10^6 M_\odot$, the mass of the orbiting object is $10M_\odot$, the spin parameter of the central BH is $S/M^2 = 0.01$, the eccentricity at the last stable orbit $e_{\text{LSO}} = 0.3$, and the frequency at the last stable orbit is calculated by $f_{\text{LSO}} = c^3/(2\pi GM) ((1 - e_{\text{LSO}}^2)/(6 + 2e_{\text{LSO}}))^3$. The evolution time is $t_c = 3.14 \times 10^6$ seconds.

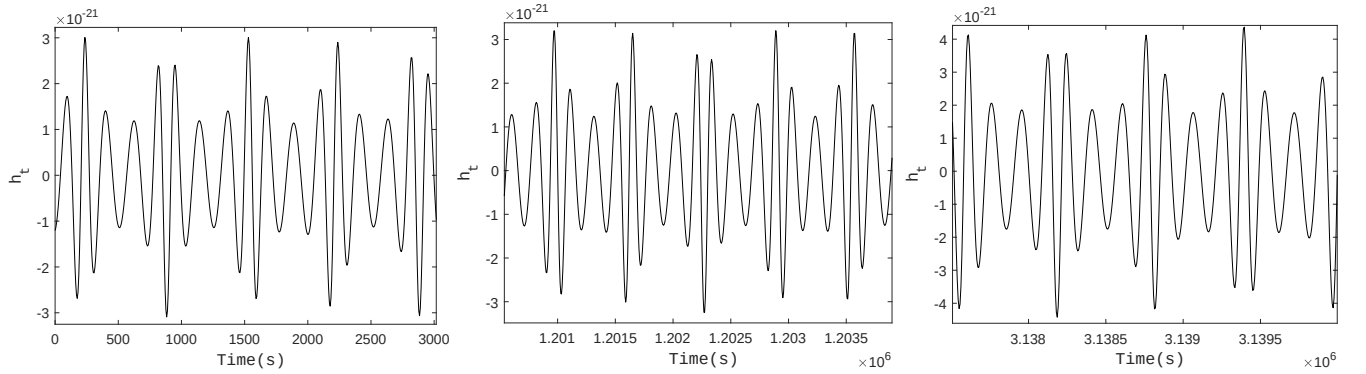


FIG. 3. The time-domain gravitational waveforms $h(t)$ are shown for three distinct time intervals: $0 \sim 3000$ (left panel), $1.109 \times 10^6 \sim 1.204 \times 10^6$ (middle panel) and $3.137 \times 10^6 \sim 3.140 \times 10^6$ (right panel). The system parameters are configured as follows: CO's mass: $m_2 = 10M_\odot$; MBH's mass: $M = 10^6 M_\odot$; MBH's spin magnitude: $S = 0.01M^2$; Angle between MBH's spin and orbital angular momentum: $\lambda = 60^\circ$; We set $\phi_{\text{LSO}} = \gamma_{\text{LSO}} = \alpha_{\text{LSO}} = 0$; Sky angle $\theta_S = \phi_S = \theta_k = \phi_k = 60^\circ$.

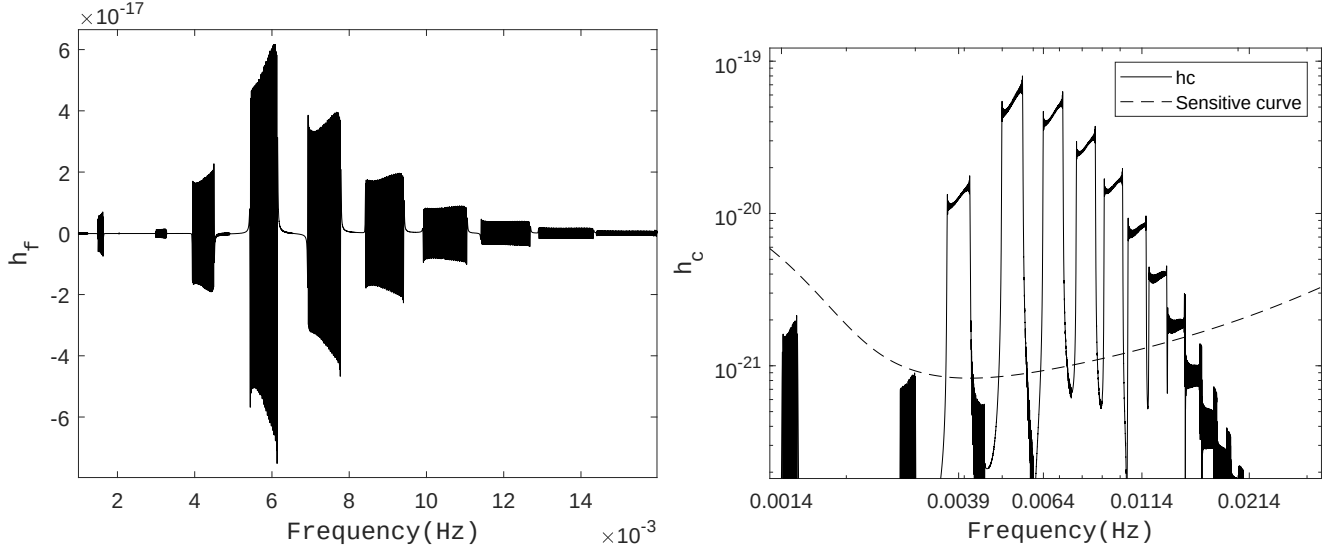


FIG. 4. GW signal on frequency domain. The frequency-domain GW waveforms by FFT on time-domain GW in Fig. 3 (Left); The characteristic strain of GW signal by $h_c = 2f|h(f)|$ (Right). The parameters are the same as in Fig. 3.

TABLE II. 68% central intervals and estimated Potential Scale Reduction Factors for four scalar summaries of the multivariate normal distribution simulated using a Metropolis algorithm. Displayed are inferences from the second halves of nine parallel sequences, stopping after 2000, 5000, and 10000 iterations. The intervals for (∞) are taken from the known normal distributions for these summaries in the target distribution.

Iteration	m_1, R	m_2, R	e_{LSO}, R	S, R
FIM	$[-1.09, 1.08] \times 10^2$	$[-1.68, 1.67] \times 10^{-3}$	$[-7.39, 7.47] \times 10^{-5}$	$[-2.48, 2.47] \times 10^{-4}$,
2000	$[-3.48, 3.89] \times 10^2, 1.828$	$[-5.48, 6.16] \times 10^{-3}, 1.832$	$[-2.68, 2.39] \times 10^{-4}, 1.824$	$[-8.15, 9.13] \times 10^{-4}, 1.827$
5000	$[-3.52, 3.13] \times 10^2, 1.146$	$[-5.51, 5.45] \times 10^{-2}, 1.146$	$[-2.13, 2.44] \times 10^{-4}, 1.146$	$[-8.32, 8.53] \times 10^{-4}, 1.146$
10000	$[-3.59, 2.81] \times 10^2, 1.046$	$[-5.66, 4.44] \times 10^{-3}, 1.045$	$[-1.94, 2.47] \times 10^{-4}, 1.046$	$[-8.50, 6.69] \times 10^{-4}, 1.046$

V. CONCLUSION

We have developed a simple and user-friendly MATLAB-based code package for Bayesian analysis of GW parameters for binaries with arbitrary eccentricities, **MatBYIB**. This package is based on the AK waveform, which employs the post-Newtonian approximation to simulate the orbital parameter evolution and utilizes the Peters-Mathews formalism [39] to obtain the GW quadrupole moment. Then, FIM and MCMC are both employed to reconstruct the posterior distributions of GW parameters. **MatBYIB** incorporates parallelization, enabling routine numerical simulations of PE on modest hardware (such as a desktop computer) without compromising numerical convergence. We demonstrated its potential applications through examples of GW parameter estimation for EMRI binaries in elliptical orbits. Comparisons between results obtained from the FIM and

those from MCMC show good agreement.

Given the excellent scalability of the AK waveform, future developments of **MatBYIB** could include additional effects, such as dynamical friction from dark matter[12, 56]. To enhance the accuracy of the waveform, higher-order post-Newtonian terms could be incorporated, or alternative waveform models such as **FastEMRI** [57] could be adopted.

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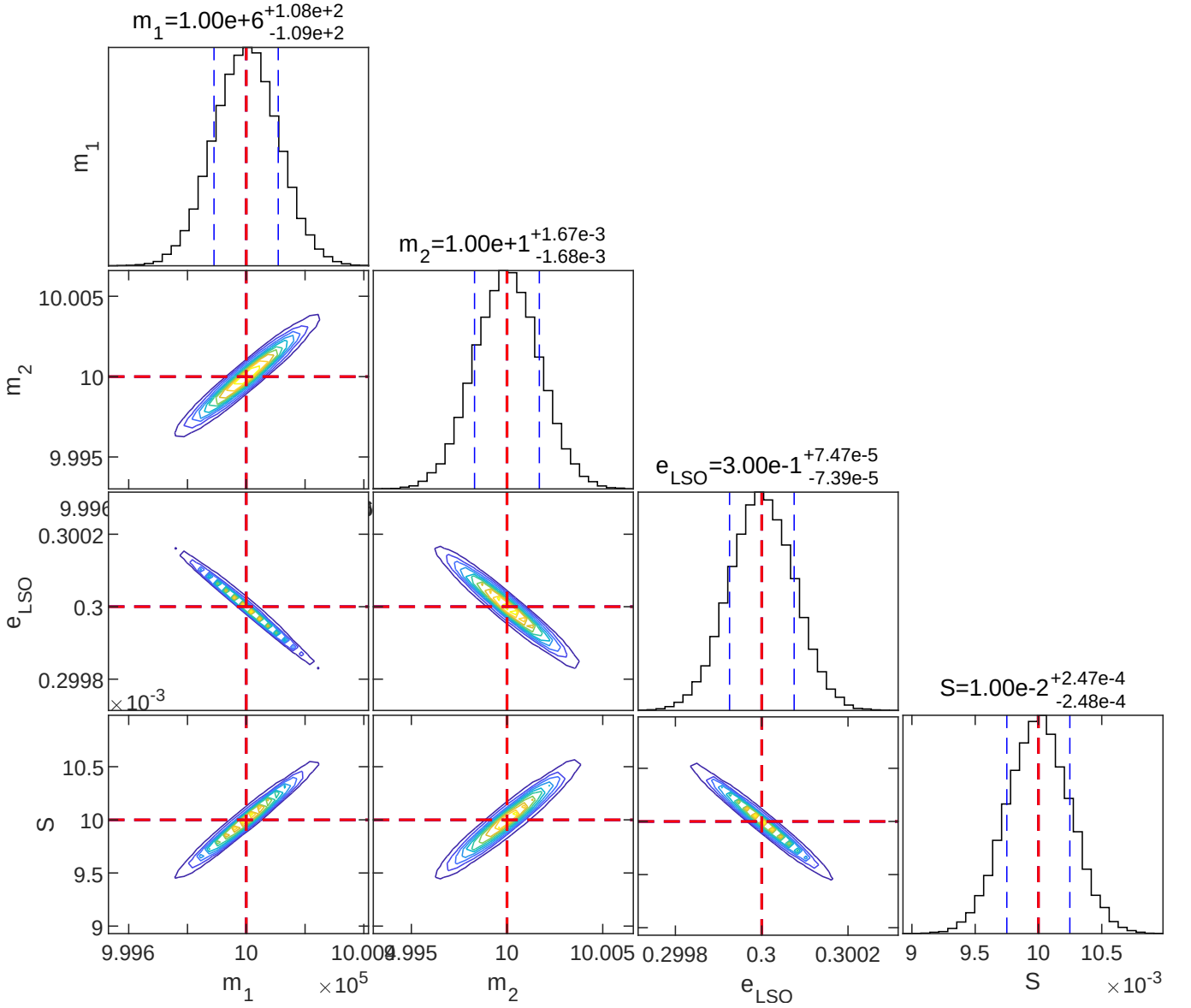


FIG. 5. The corner plot from FIM exploration with fiducial/injected values $m_1 = 10^6 M_\odot$ (central mass), $m_2 = 10 M_\odot$ (orbiting mass), $S/M^2 = 0.01$ (dimensionless spin of the central BH), $e_{\text{LSO}} = 0.3$. We have assumed an observation of $t_c = 3.14 \times 10^6 \text{ s}$, redshift $z = 0.01$, Median and 68% confidence interval are $m_1 = 1^{+1.08 \times 10^{+2}}_{-1.09 \times 10^{+2}} \times 10^6 M_\odot$, $m_2 = 10^{+1.69 \times 10^{-3}}_{-1.70 \times 10^{-3}} M_\odot$, $e_{\text{LSO}} = 0.3^{+7.49 \times 10^{-5}}_{-7.39 \times 10^{-5}}$, and $S/M^2 = 0.01^{+2.47 \times 10^{-4}}_{-2.49 \times 10^{-4}}$.

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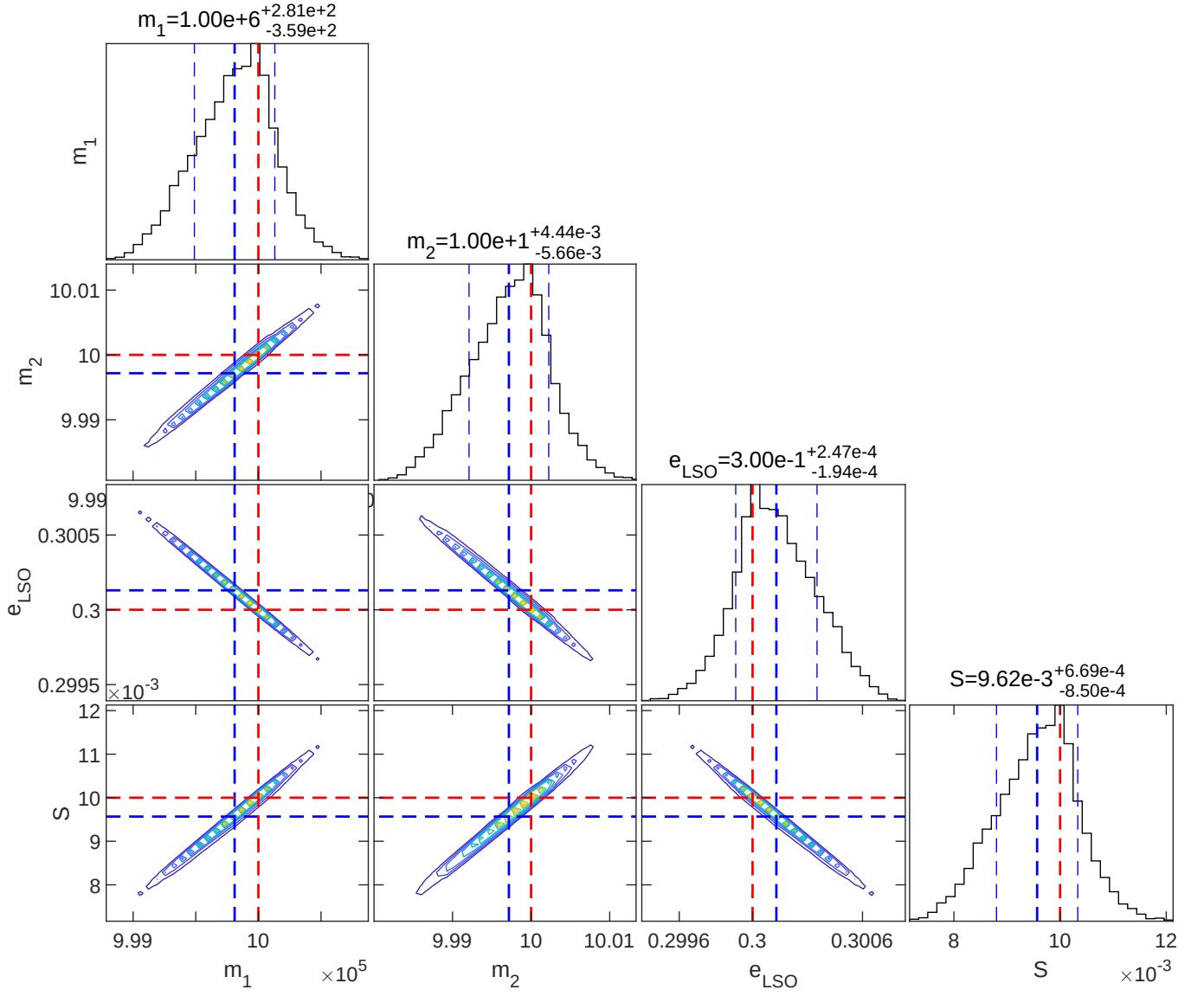


FIG. 6. The corner plot from MCMC exploration with binary system whose parameters are the same as Fig. 5, Median and 68% confidence interval are $m_1 = 1^{+2.81 \times 10^2}_{-3.59 \times 10^2} \times 10^6 M_\odot$, $m_2 = 10^{+4.44 \times 10^{-3}}_{-5.65 \times 10^{-3}} M_\odot$, $e_{\text{LSO}} = 0.3^{+2.47 \times 10^{-4}}_{-1.94 \times 10^{-4}}$ and $S/M^2 = 9.62^{+6.69 \times 10^{-4}}_{-8.50 \times 10^{-3}} \times 10^{-3}$.

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